Decentralized Ad Network with Smart Contracts and ML-Based Fraud Detection

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***Abstract*—Digital advertising has been dominated by many centralized platforms that suffer from limited transparency, high intermediate costs with increasing risk of ad fraud. Advertisers are often unaware how impressions and clicks are verified due to lack of visibility, while publishers face delayed revenue distribution. To address these challenges, this paper proposes a Decentralized Advertising Network that combines blockchain-based smart contracts along with machine learning based fraud detection, thereby creating a transparent, trustless, fraud-resistant ecosystem. In this proposed approach, the advertisement assets are stored on Inter Planetary File System (IPFS) in order to ensure tamper-free and decentralized storage. Smart Contracts are deployed on a blockchain network to automate campaign registration, validation of interactions and publisher payments without relying on centralized systems. Experiment results have demonstrated that combining blockchain transparency with ML-based validation significantly reduces fraudulent payouts. The proposed framework presents a scalable and secure alternative to traditional digital advertising networks.**

***Index Terms*—Blockchain, Decentralized Advertising, Smart Contracts, Machine Learning, Click Fraud Detection, IPFS, Fraud Prevention, Digital Advertising Security.**

I. Introduction

Online advertising has become a complicated and profitable system that supports many digital platforms and services. Advertisers need ad networks to provide measurable engagement, while publishers count on these networks for steady income. However, this relationship depends on centralized middlemen who manage the current digital advertising setup. This structure leads to problems with transparency, accountability, and trust. Stakeholders frequently do not have a clear view of how impressions, clicks, and payments are handled. This lack of visibility can cause disputes and inefficiencies throughout the advertising process.

The battle with and against fake news in this environment is an ongoing game of whack-a-mole. Bots, fake traffic sources and malicious scripts generate fraudulent impressions and clicks. This distorts the picture of how a campaign is performing, and inflates costs to advertise. Click fraud results in monetary losses for advertisers and undermines the confidence in digital ad exchanges. Static rule-based traditional fraud detection techniques, which are usually centralized in nature, cannot adapt to emerging attack patterns and advanced organized fraudulent activities.

Lately, distributed ledger technologies have begun to solve trust issues in digital systems. Blockchain is particularly notable because it keeps records secure and unchangeable without giving control to a single authority. When you add smart contracts, you can automate deals, track transactions openly, and ensure that settlements cannot be tampered with. In digital advertising, this setup reduces dishonest practices. Campaigns operate more transparently, and payments go to the right places without the usual confusion.

However, transparency alone does not stop fraudulent behavior during interactions. Invalid impressions and clicks can still occur and be recorded unless they are spotted before being logged. This issue shows the need for a smart detection layer that can assess behavioral patterns instead of relying on fixed rules. Machine Learning (ML) techniques offer a way to identify unusual interactions by learning from past traffic patterns and setting apart real user behavior from fraudulent activity.

In this paper, we introduce a decentralized advertising network that brings together blockchain smart contracts and machine learning for fraud detection. The framework takes ad management and payment settlement out of the hands of a single authority. It leans on smart models to check ad interactions as they happen, catching fraud before anything gets logged on the blockchain. That way, only verified interactions shape campaign analytics and payments. By combining built-

in trust with adaptable fraud detection, this system addresses the transparency and security problems that plague today's ad networks.

The proposed approach aims to reduce dependency on centralized intermediaries, improve confidence among advertisers and publishers, and enhance the overall reliability of digital advertising operations. Through this integration of decentralization and intelligent analysis, the work contributes toward the development of a more accountable and resilient advertising ecosystem.

II. Related Work

Ever since digital advertising was introduced it came along with issues of fraud,lack of transparency and reliance on centralized systems. Recent researches has explored the use of blockchain and machine together to address these challenges.Several studies had focused on machine learning based click fraud detection. Alzahrani et al demonstrated that tree based models such as Random Forest,gradient Boosting along with deep learning models like RNN and CNN can effectively distinguish fraudulent clicks from legitimate user interactions making use of behavioral and temporal features [1]. This work highlights importance of feature engineering and model selection in detecting patterns of fraud.Researchers have also explored blockchain technology for improving advertising transparency. Walia et al suggested use of decentralized ledgers inorder to verify ad impressions and clicks thereby ensuring advertising events are tamper-proof [2]. Gujarathi et al discussed how using blockchain can enhance trust and reduce fraud in digital marketing by enabling transparency in transactions [3]. While these works emphasize transparency and accountability, they do not incorporate real-time fraud detection mechanisms. Liu et al introduced smart advertising network based on blockchain that uses cryptographic proofs ensuring transparency and protecting user privacy as well [4]. However, framework majorly focuses on verification than proactive detection of fraudulent transactions in the first place.

More broadly, studies on Web 3.0 and decentralized applications have highlighted the benefits of trustless systems and smart contract automation. The studies emphasized decentralized control, transparency and peer-to-peer interactions for internet applications [5]. These principles agree with decentralized advertising ecosystem but do not address fraud detection.

Despite these advancements, a research gap still exists. Existing ML-based detection systems are centralized and lack transparency, verifiable payment mechanisms, while blockchain-based advertising solutions emphasize transparency but lack intelligent, real-time fraud detection. To address this gap, the proposed system integrates real time ML fraud detection with smart contract-based ad payment enforcement and IPFS-based decentralized storage thereby creating a transparent, automated and fraud-resistant advertising network. This overall hybrid approach ensures that only ML-verified interactions are recorded and used to trigger payments and bridging intelligent fraud detection and decentralized trust mechanisms.

III. Proposed Methodology

The proposed system adopts a hybrid decentralized architecture that combines blockchain-based smart contracts with machine learning–driven fraud detection to address transparency and security challenges in digital advertising. The methodology is designed to ensure that only legitimate ad interactions are rewarded while fraudulent activity is detected and filtered before financial settlement occurs.

The overall workflow is divided into four logical layers: ad interaction collection, fraud analysis, decentralized validation and settlement, and analytics and storage. Each layer is implemented as an independent module to maintain modularity, scalability, and ease of maintenance.

A. Ad Interaction Collection

Ad interactions are initiated at the publisher's website through a lightweight client-side script embedded within the web page. This script is responsible for rendering the assigned advertisement and monitoring user interaction events such as clicks. Upon interaction, the script captures contextual metadata including IP address, user-agent string, screen parameters, and timestamp information. These data points are transmitted to the backend server through a secure API endpoint.

The backend server, implemented using a Node.js and Express framework, acts as the coordination layer of the system. Incoming requests are first validated to ensure that the publisher is authorized and that the ad campaign is legitimately booked. This initial verification prevents unauthorized or malformed requests from entering subsequent stages of processing.

B. Machine Learning–Based Fraud Detection

Once basic validation is completed, click data is forwarded to a dedicated fraud detection service. This service extracts and constructs behavioral features such as time intervals between consecutive clicks, repetition patterns originating from the same IP address, and user-agent consistency. These features are designed to capture temporal and behavioral anomalies commonly associated with automated or malicious traffic.

The prepared feature vector is submitted to a machine learning model hosted as a separate service. The model performs real-time inference and classifies each interaction as either valid or fraudulent. By relying on learned behavioral patterns rather than static rules, the system adapts more effectively to evolving fraud strategies. The classification result is returned to the backend, where it directly influences the transaction decision.

C. Smart Contract–Based Validation and Settlement

For interactions classified as valid, the backend initiates a blockchain transaction through a smart contract deployed on an Ethereum-compatible network. The smart contract encodes the rules governing publisher registration, campaign participation, and payment settlement. Upon execution, the contract

credits the publisher in a transparent and tamper-resistant manner, ensuring that payments are automated and verifiable.

Interactions identified as fraudulent are excluded from on-chain recording, preventing invalid data from being permanently stored on the blockchain. This selective recording approach preserves the integrity of the ledger while avoiding unnecessary transaction costs associated with fraudulent activity. From an economic perspective, selective on-chain recording minimizes gas consumption by avoiding unnecessary writes for fraudulent interactions. This optimization reduces operational costs while maintaining ledger integrity. By ensuring that only ML-verified interactions influence financial settlement, the architecture achieves both cost efficiency and fairness within the decentralized advertising ecosystem.

D. Decentralized Storage

Advertisement assets such as images and media are stored via the InterPlanetary File System (IPFS), which, instead of traditional location-based storage, utilizes content-based addressing. When an advertiser adds an asset, the IPFS generates a unique identifier known as the Content Identifier (CID) based on the SHA-256 cryptographic hash of the content. This CID can be considered a digital fingerprint for the asset. This method ensures that if the advertiser tries to change the asset, which can be considered a form of "bait and switch" fraud, the CID will be different from the new one generated for the new asset, ensuring that the integrity of the content.

E. System Integration

The end-to-end processing of an ad interaction follows a sequential API-driven workflow, ensuring modularity and secure communication across components. When a user clicks on an advertisement displayed on the publisher's website, the embedded client-side script immediately sends an HTTP request to the backend server. This request contains contextual interaction data, including campaign ID, publisher ID, timestamp, device metadata, and network attributes. The Node.js backend acts as the central coordinator of the workflow. Upon receiving the interaction request, the backend performs preliminary validation and then forwards the extracted feature set to the machine learning microservice. This request is sent to the Flask-based ML service, which performs real-time inference using the deployed fraud detection model. The service responds with a classification result (valid or fraudulent) along with an associated risk score.If the interaction is classified as valid, the backend initiates a blockchain transaction by invoking the smart contract through Ethers.js. If the interaction is flagged as fraudulent, the transaction is blocked and logged in the analytics system without triggering a blockchain update.

F. Algorithmic Representation

The end-to-end process of handling an ad interaction is described in Algorithm 1.

Algorithm 1 Ad Interaction Validation and Settlement

Require: Ad interaction data D (IP, user-agent, timestamp, publisher, campaign)
Ensure: Valid interaction settlement or fraud rejection

```
Receive interaction data D from publisher-side script
Verify publisher registration and campaign authorization
if authorization fails then
    Reject interaction and terminate process
end if
Extract behavioral features from D
Send feature vector to ML fraud detection service
Receive fraud classification result R
if R = Fraudulent then
    Log interaction as fraud in analytics module
    Block blockchain transaction
else
    Invoke smart contract to record valid interaction
    Credit publisher via blockchain settlement
    Log interaction as valid in analytics module
end if
End process
```

G. Analytics Dashboard

There is a feature of real-time analytics that is designed to improve transparency for the advertisers and publishers who uses this system. It offers a comparative view of total interaction requests received from the publisher sites and ML-approved interactions that triggered blockchain transactions.

The feature is termed the Transparency Dashboard, which allows the advertisers to view the number of fraudulent interactions filtered out by the ML verification pipeline.

H. End-to-End Workflow Summary

The proposed methodology ensures that ad interactions undergo multi-stage verification before financial settlement. By integrating machine learning–based fraud detection prior to blockchain recording, the system prevents fraudulent behavior from influencing both analytics and payments. The decentralized execution of settlement logic through smart contracts further eliminates reliance on centralized intermediaries, resulting in a secure, transparent, and accountable advertising framework.

I. Novelty of the Proposed Framework

The major novelty about the proposed framework is that it combines real-time machine learning validation with settlement using blockchain technology in one unified workflow. Previous studies have looked at using blockchain to make things more transparent or using machine learning to detect fraud. This system uses both in a way that relies on each other. This means that the smart validation process directly controls how transactions are recorded in a way that cannot be changed.

The proposed system is different from decentralized advertising models because it checks things before they are

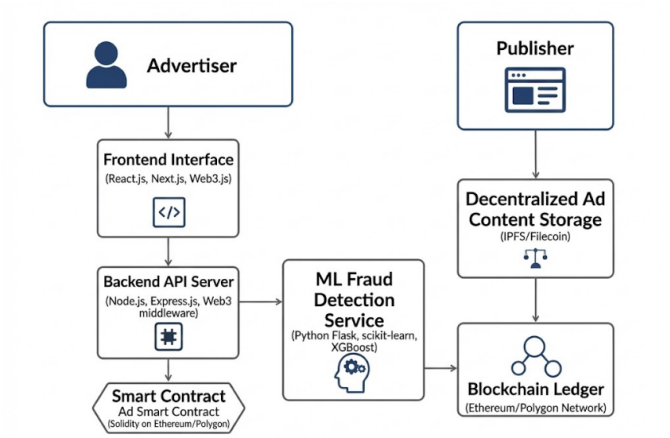

Fig. 1. Proposed Decentralized Advertising Network Architecture integrating blockchain, IPFS storage and ML-based fraud detection.

finalized. It does this by introducing a step where things are validated before the blockchain is used. This means that fake interactions are stopped before they can be recorded permanently or trigger payments. This approach helps keep the ledger accurate and makes transactions more cost-effective.

Overall what is new about this work is not the use of blockchain or machine learning but how they are used together. This approach creates a system for advertising settlements that's aware of fraud and minimizes the need for trust. The proposed framework is a step forward because it uses machine learning and blockchain technology in a coordinated way. The machine learning validation and blockchain technology work together to create a system that's fair and trustworthy. The framework is designed to prevent fraud and make sure that transactions are accurate. This is an improvement, over other systems that do not use machine learning and blockchain technology together.

IV. Conclusion and Future Work

This paper proposed a decentralized advertising framework that integrates blockchain-based smart contracts with machine learning–driven fraud detection to address transparency and click fraud challenges in traditional advertising networks. By introducing a real-time ML validation layer before blockchain settlement, the system ensures that only legitimate interactions are recorded and rewarded. Smart contracts automate payment execution in a transparent and tamper-resistant manner, while decentralized storage enhances data integrity. The combined approach improves trust, reduces fraudulent payouts, and eliminates reliance on centralized intermediaries. The transparency feature helps create trust among the stakeholders by showing measurable data regarding fraud mitigation and making sure that all transactions are tied to verified engagements. The feature of decentralized storage and cryptography helps improve the security of the overall advertising network.

Although the prototype demonstrates feasibility and effectiveness, further improvements are required for large-scale deployment. Future work includes integrating Layer-2 blockchain solutions to reduce transaction costs, enhancing the fraud detection model using larger real-world datasets and adaptive learning techniques, and evaluating system performance under high-traffic conditions. Incorporating privacy-preserving mechanisms and formal smart contract auditing may also strengthen production readiness. With these enhancements, the framework can evolve into a scalable and secure decentralized advertising infrastructure.

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